In [29]:
```
# --- Imports ---
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import unicodedata
import re
import datetime
import networkx as nx
from itertools import combinations
import pycountry
import plotly.graph_objects as go
```

Loading and Cleaning

In [30]:
```
# --- Data Loading & Cleaning for Flattened OpenAlex ---
def load_and_clean_your_csv(file_path):
    """
    Cleans the flattened OpenAlex CSV structure found in your research01 file.
    """
    df = pd.read_csv(file_path)

    # 1. Standardize Numeric Columns
    df['publication_year'] = pd.to_numeric(df['publication_year'], errors='coe
    df['cited_by_count'] = pd.to_numeric(df['cited_by_count'], errors='coerce'

    # 2. Extract Primary Author (takes the first name before the first '|')
    def get_primary_author(name_str):
        if pd.isna(name_str): return "Unknown"
        return str(name_str).split('|')[0]

    # 3. Handle Topics/Keywords (using the 'primary_topic.display_name' column
    # Note: If you want to use the full topic string, we keep it as is.
    df['primary_author'] = df['authorships.author.display_name'].apply(get_prin

    return df

# Load your specific file
df_clean = load_and_clean_your_csv('research02_works-csv-Vda3njn3VwEnvWU2f6fE5
```

In [31]:
```
df_clean.head()
```

Out[31]:

	id	abstract	authorships.author.display_name	
0	https://openalex.org/W2070911198	NaN	Yanfang Ma|Jiuping Xu	https://openalex.org/A51(
1	https://openalex.org/W4211132698	NaN	Merve Cengiz Toklu	https://op
2	https://openalex.org/W2999440819	NaN	Sadia Samar Ali|Turan Paksoy|Belkız Torğul|Raj...	https://openalex.org/A507
3	https://openalex.org/W2254975279	NaN	Yanfang Ma|Jiuping Xu	https://openalex.org/A51(
4	https://openalex.org/W854438218	NaN	Fatemeh Torfi|Reza Zanjirani Farahani|Iraj Mah...	https://openalex.org/A50(

5 rows × 30 columns

In [32]:

```
# --- Rigorous cleaning (accents and extrange characters) --- #

def rigorous_clean(text):
    if not isinstance(text, str):
        return text

    # 1. Remove the '' and non-printable artifacts
    # This encodes to ASCII and ignores anything it can't translate
    text = text.encode("ascii", "ignore").decode("ascii")

    # 2. Normalize to remove accents correctly
    nksel = unicodedata.normalize('NFKD', text)
    clean_text = "".join([c for c in nksel if not unicodedata.combining(c)])

    # 3. Use Regex to keep only letters, numbers, pipes (for your separators),
    # This removes the "weird elements" and extra spaces between letters
    clean_text = re.sub(r'[^a-zA-Z0-9\s|.\-]', '', clean_text)

    # 4. Collapse multiple spaces into one
    clean_text = re.sub(r'\s+', ' ', clean_text).strip()

    return clean_text

# Apply to your columns
cols_to_fix = ['authorships.author.display_name', 'display_name', 'primary_topi

for col in cols_to_fix:
    if col in df_clean.columns:
        df_clean[col] = df_clean[col].apply(rigorous_clean)

print("Rigorous cleaning finished. No more squares or extra spaces.")
```

```
Rigorous cleaning finished. No more squares or extra spaces.
```

In [33]:

```
df_clean.head()
```

Out[33]:

	id	abstract	authorships.author.display_name	
0	https://openalex.org/W2070911198	NaN	Yanfang Ma|Jiuping Xu	https://openalex.org/A51(
1	https://openalex.org/W4211132698	NaN	Merve Cengiz Toklu	https://op
2	https://openalex.org/W2999440819	NaN	Sadia Samar Ali|Turan Paksoy|Belkz Torul|Rajbi...	https://openalex.org/A507
3	https://openalex.org/W2254975279	NaN	Yanfang Ma|Jiuping Xu	https://openalex.org/A51(
4	https://openalex.org/W854438218	NaN	Fatemeh Torfi|Reza Zanjirani Farahani|Iraj Mah...	https://openalex.org/A50(

5 rows × 30 columns

Function Creation

In [34]:

```
# --- Production Over Time ---
def plot_production_trends(df):
    # Filter years to match your study range (e.g., 2015-2025)
    yearly_counts = df[df['publication_year'] >= 2000]['publication_year'].valu

    plt.figure(figsize=(12, 6))
    sns.barplot(x=yearly_counts.index.astype(int), y=yearly_counts.values, colo
    plt.title('Documents per Year (VRP Review)', fontsize=16)
    plt.xlabel('Year', fontsize=14)
    plt.ylabel('Number of Documents', fontsize=14)
    plt.xticks(fontsize=12) # Increases size of the numbers on the X-axis
    plt.yticks(fontsize=12) # Increases size of the Author Names on the Y-axis
    plt.xticks(rotation=45)
    plt.grid(axis='y', linestyle='--', alpha=0.7)
    plt.show()
```

In [35]:

```
## -- Get country name ---##
def get_country_name(code):
    try:
        # Translates 'MX' -> 'Mexico', 'CN' -> 'China', etc.
        return pycountry.countries.get(alpha_2=code).name
    except:
        return code # Returns original code if translation fails
```

In [36]:

```
# --- Filtered Topic Analysis ---
def plot_real_trends(df):
    """
    Excludes the broad search terms to see the underlying application
    areas for Hybrid Augmented Intelligence.
    """
    # 1. Define the terms to remove (VRP and its obvious synonyms)
    exclude_list = [
        'vehicle routing problem',
        'vrp',
        'vehicle routing optimization methods',
        'routing optimization'
```

```
    ]

    # 2. Get the value counts of the primary topic column
    topic_counts = df['primary_topic.display_name'].value_counts()

    # 3. Filter the series using a negative regex match
    # This keeps only topics that DO NOT contain our exclusion terms
    pattern = '|'.join(exclude_list)
    filtered_topics = topic_counts[~topic_counts.index.str.contains(pattern, ca

    # 4. Grab the top 10 for the visualization
    top_10_real = filtered_topics.head(10)

    # 5. Visualization
    plt.figure(figsize=(15, 6))
    sns.barplot(x=top_10_real.values, y=top_10_real.index, palette='magma')
    plt.title('Top 10 Research Domains (VRP Terms Removed)', fontsize=14)
    plt.xlabel('Number of Articles')
    plt.ylabel('')
    plt.xticks(fontsize=12) # Increases size of the numbers on the X-axis
    plt.yticks(fontsize=12) # Increases size of the Author Names on the Y-axis
    plt.tight_layout()
    plt.show()
```

In [37]:

```
# --- Top Cited Articles (Identification of Influential Works) ---
def show_top_cited(df):
    """Shows the most influential papers for your review."""
    top_papers = df[['display_name', 'primary_author', 'publication_year', 'cit
    return top_papers
```

In [38]:

```
##--- Plot top stats ----

def plot_top_stats(df, category='author', metric='publications', top_n=10,
                   citation_col='cited_by_count', separator='|'):
    """
    Unified function adjusted for your specific research01 dataset column names
    """

    # 1. Map generic categories to YOUR specific DataFrame column names
    column_map = {
        'author': 'authorships.author.display_name',
        'institution': 'authorships.institutions.display_name',
        'country': 'authorships.countries'
    }

    target_col = column_map.get(category.lower())

    if not target_col or target_col not in df.columns:
        print(f"Error: Category '{category}' invalid or column '{target_col}' n
        return

    # 2. Pre-process: Copy only needed columns
    cols_needed = [target_col]
    if metric == 'citations':
        cols_needed.append(citation_col)
    df_temp = df[cols_needed].copy()

    # 3. Clean and Explode (Split values by the pipe '|')
    df_temp[target_col] = df_temp[target_col].fillna('').astype(str).str.split(
```

```
    exploded = df_temp.explode(target_col)

    # 4. Rigorous Cleaning
    if category == 'author':
        exploded[target_col] = exploded[target_col].apply(rigorous_clean)
    elif category == 'country':
        # Translate codes to names
        exploded[target_col] = exploded[target_col].str.strip().apply(get_count
    else:
        exploded[target_col] = exploded[target_col].str.strip()

    # Remove garbage strings (length <= 1)
    exploded = exploded[exploded[target_col].str.len() > 1]

    # 5. Calculation Logic
    if metric == 'publications':
        top_data = exploded.groupby(target_col).size().sort_values(ascending=Fa
        title_text = f'Top {top_n} {category.capitalize()}s (by Publication Cou
        x_label = 'Number of Documents'
        palette = 'viridis'

    elif metric == 'citations':
        top_data = exploded.groupby(target_col)[citation_col].sum().sort_values
        title_text = f'Top {top_n} {category.capitalize()}s (by Total Citations
        x_label = 'Total Citations'
        palette = 'rocket'

    # 6. Visualization
    plt.figure(figsize=(12, 8))
    sns.barplot(x=top_data.values, y=top_data.index, palette=palette)

    plt.title(title_text, fontsize=16)
    plt.xlabel(x_label, fontsize=14)
    #plt.ylabel(category.capitalize(), fontsize=14)
    plt.ylabel('')
    plt.xticks(fontsize=12)
    plt.yticks(fontsize=12)
    plt.tight_layout()
    plt.show()

    return top_data
```

In [39]:

```
# --- Filtered database by the last n years ----#
def filter_by_last_n_years(df, n_years):
    """
    Filters the dataframe to include only articles from the last n_years.
    Useful for defining the '2015-2025' timeframe for your review.
    """
    # 1. Ensure publication_year is numeric
    df['publication_year'] = pd.to_numeric(df['publication_year'], errors='coer

    # 2. Determine the reference year (Current year is 2026)
    current_year = 2026
    start_year = current_year - n_years

    # 3. Create the filtered dataframe
    filtered_df = df[df['publication_year'] >= start_year].copy()

    print(f"Original articles: {len(df)}")
```

```
    print(f"Articles from {start_year} to {current_year}: {len(filtered_df)}")

    return filtered_df
```

In [40]:

```
## --- Author clusters ----#
def analyze_author_clusters(df, authors_column, min_collaborations=4, separato
                            grid_rows=2, grid_cols=2, node_size=500, font_size=
                            edge_width_scale=0.5):
    """
    Plots the top author clusters in a grid layout.

    Parameters:
    - df: DataFrame containing the data.
    - authors_column: Name of the column with authors.
    - min_collaborations: Minimum shared papers to create a link.
    - separator: Character separating authors (e.g., '|' or ';').
    - grid_rows, grid_cols: Dimensions of the subplot grid (e.g., 2x2 = top 4 c
    - node_size: Visual size of the author bubbles.
    - font_size: Size of the author names.
    - edge_width_scale: Multiplier for edge thickness.
    """

    # --- 1. BUILD THE GRAPH ---
    G = nx.Graph()

    # Iterate through publications
    for authors_str in df[authors_column].dropna():
        # Split and clean author names
        authors = [a.strip() for a in str(authors_str).split(separator) if a.st

        # Create edges for every pair of co-authors
        if len(authors) > 1:
            for u, v in combinations(authors, 2):
                if G.has_edge(u, v):
                    G[u][v]['weight'] += 1
                else:
                    G.add_edge(u, v, weight=1)

    # --- 2. FILTER & SORT CLUSTERS ---
    # Keep only edges that meet the minimum collaboration threshold
    strong_edges = [(u, v) for u, v, d in G.edges(data=True) if d['weight'] >=
    G_filtered = G.edge_subgraph(strong_edges).copy()

    # Get all connected components (clusters) and sort them by size (largest fi
    # This ensures we prioritize the "most collaborative" groups
    clusters = sorted(nx.connected_components(G_filtered), key=len, reverse=Tru

    if not clusters:
        print(f"No collaborations found with >= {min_collaborations} shared pap
        return

    # --- 3. CREATE THE GRID PLOT ---
    num_plots = grid_rows * grid_cols
    fig, axes = plt.subplots(grid_rows, grid_cols, figsize=(grid_cols * 6, gric

    # Flatten axes array for easy iteration (handles 1x1, 1xN, NxN cases)
    if grid_rows * grid_cols > 1:
        axes = axes.flatten()
    else:
```

```
        axes = [axes]

    print(f"Found {len(clusters)} clusters. Showing the top {min(len(clusters),

    for i in range(num_plots):
        ax = axes[i]

        # If we run out of clusters, hide the empty subplot
        if i >= len(clusters):
            ax.axis('off')
            continue

        # Get the subgraph for this specific cluster
        cluster_nodes = clusters[i]
        sub_G = G_filtered.subgraph(cluster_nodes)

        # Layout: distinct shape for each cluster
        pos = nx.spring_layout(sub_G, k=0.5, seed=42)

        # Dynamic edge thickness based on collaboration count
        weights = [sub_G[u][v]['weight'] for u, v in sub_G.edges()]
        widths = [w * edge_width_scale for w in weights]

        # Draw the plot
        nx.draw_networkx_nodes(sub_G, pos, ax=ax, node_size=node_size, node_col
        nx.draw_networkx_edges(sub_G, pos, ax=ax, width=widths, alpha=0.5, edge
        nx.draw_networkx_labels(sub_G, pos, ax=ax, font_size=font_size, font_fa

        ax.set_title(f"Cluster #{i+1} ({len(cluster_nodes)} Authors)", fontsize
        ax.axis('off')

    plt.tight_layout()
    plt.show()

# --- EXAMPLE USAGE ---
# analyze_author_clusters(
#     df,
#     authors_column='Authors',
#     min_collaborations=5,
#     separator='|',
#     grid_rows=2,
#     grid_cols=2,   # This will show the top 4 biggest groups
#     node_size=600,
#     font_size=8
# )
```

In [41]:

```
## --- Colaboration clusters --- ##
def analyze_category_clusters(df, category='author', min_shared=4,
                              grid_rows=2, grid_cols=2, node_size=20,
                              font_size=12, edge_width_scale=0.5):
    """
    Generalized function to plot collaboration clusters for Authors, Institutio
    """

    # 1. Map categories to your specific CSV columns
    column_map = {
        'author': 'authorships.author.display_name',
        'institution': 'authorships.institutions.display_name',
        'country': 'authorships.countries'
```

```
    }

    target_col = column_map.get(category.lower())
    if not target_col:
        print(f"Error: Category '{category}' not supported.")
        return

    # 2. Country name translation helper
    def translate_country(code):
        try: return pycountry.countries.get(alpha_2=code.strip()).name
        except: return code.strip()

    # 3. Build the Graph
    G = nx.Graph()

    for items_str in df[target_col].dropna():
        # Split and clean
        items = [i.strip() for i in str(items_str).split('|') if i.strip()]

        # Category-specific cleaning
        if category == 'author':
            items = [rigorous_clean(i) for i in items]
        elif category == 'country':
            items = [translate_country(i) for i in items]

        # Get UNIQUE items per paper to find true COLLABORATION (e.g. Universi
        unique_items = list(set(items))

        if len(unique_items) > 1:
            for u, v in combinations(unique_items, 2):
                if G.has_edge(u, v):
                    G[u][v]['weight'] += 1
                else:
                    G.add_edge(u, v, weight=1)

    # 4. Filter strong connections & find clusters
    strong_edges = [(u, v) for u, v, d in G.edges(data=True) if d['weight'] >=
    G_filtered = G.edge_subgraph(strong_edges).copy()
    clusters = sorted(nx.connected_components(G_filtered), key=len, reverse=Tru

    if not clusters:
        print(f"No {category} clusters found with >= {min_shared} shared papers
        return

    # 5. Plot Grid
    num_plots = grid_rows * grid_cols
    fig, axes = plt.subplots(grid_rows, grid_cols, figsize=(grid_cols * 6, gric
    axes = axes.flatten() if num_plots > 1 else [axes]

    print(f"Showing top {min(len(clusters), num_plots)} {category} clusters.")

    for i in range(num_plots):
        ax = axes[i]
        if i >= len(clusters):
            ax.axis('off')
            continue

        sub_G = G_filtered.subgraph(clusters[i])
        pos = nx.spring_layout(sub_G, k=0.6, seed=42)
```

```
        # Scale edge width by weight
        widths = [sub_G[u][v]['weight'] * edge_width_scale for u, v in sub_G.ec

        nx.draw_networkx_nodes(sub_G, pos, ax=ax, node_size=node_size, node_col
        nx.draw_networkx_edges(sub_G, pos, ax=ax, width=widths, alpha=0.4, edge

        # Dynamic label positioning
        labels = {node: node.replace(' ', '\n') for node in sub_G.nodes()}
        nx.draw_networkx_labels(sub_G, pos, labels=labels, ax=ax, font_size=for

        ax.set_title(f"{category.capitalize()} Cluster #{i+1} ({len(clusters[i]
        ax.axis('off')

    plt.tight_layout()
    plt.show()
```

In [42]:

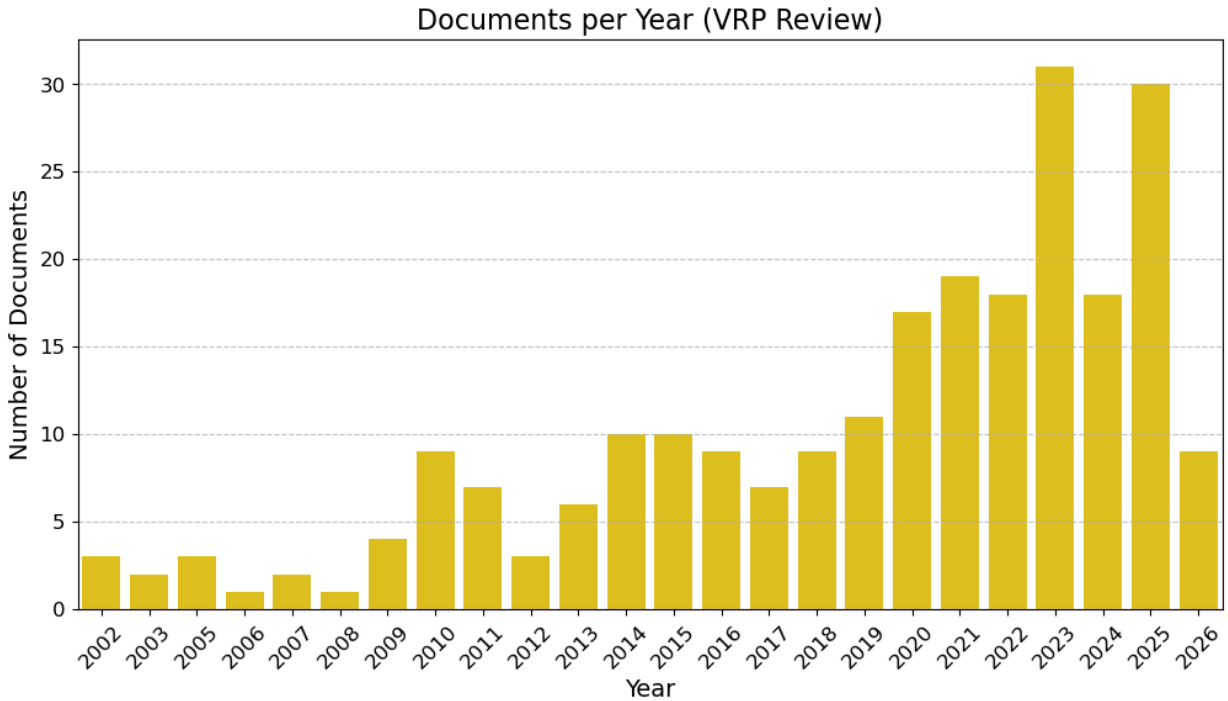
```
## --- Sankey diagrams ---##

def plot_sankey_categories(df, categories=['country', 'topic', 'author'], n_ite
    # 1. Column Mapping (Specific to your research01.csv)
    column_map = {
        'author': 'authorships.author.display_name',
        'institution': 'authorships.institutions.display_name',
        'country': 'authorships.countries',
        'topic': 'primary_topic.display_name'
    }

    # 2. Filtering Logic (VRP Exclusions)
    exclude_vrp = ['vehicle routing problem', 'vrp', 'optimization methods', 'i
    pattern = '|'.join(exclude_vrp)

    def get_country_name(code):
        try: return pycountry.countries.get(alpha_2=code).name
        except: return code

    # 3. Processing
    cols = [column_map[c] for c in categories]
    temp_df = df[cols].dropna().copy()

    for cat in categories:
        col = column_map[cat]
        temp_df[col] = temp_df[col].astype(str).str.split('|')
        temp_df = temp_df.explode(col)
        temp_df[col] = temp_df[col].str.strip()

        if cat == 'author':
            temp_df[col] = temp_df[col].apply(rigorous_clean)
        elif cat == 'country':
            temp_df[col] = temp_df[col].apply(get_country_name)
        elif cat == 'topic':
            temp_df = temp_df[~temp_df[col].str.contains(pattern, case=False, r
            # TRUNCATE LONG TOPICS for visual clarity
            temp_df[col] = temp_df[col].apply(lambda x: (x[:25] + '...') if len

    # Filter for top N to avoid "horrible" clutter
    for cat in categories:
        col = column_map[cat]
        top_items = temp_df[col].value_counts().head(n_items).index
        temp_df = temp_df[temp_df[col].isin(top_items)]
```

```
    # 4. Create Links
    links = []
    for i in range(len(categories) - 1):
        source_col = column_map[categories[i]]
        target_col = column_map[categories[i+1]]
        res = temp_df.groupby([source_col, target_col]).size().reset_index(name
        res.columns = ['source', 'target', 'value']
        links.append(res)

    sankey_df = pd.concat(links)
    all_nodes = list(pd.concat([sankey_df['source'], sankey_df['target']]).unic
    node_map = {node: i for i, node in enumerate(all_nodes)}

    # 5. Build the Chart with specific "Clean" styling
    fig = go.Figure(data=[go.Sankey(
        node=dict(
            pad=30,           # More space between boxes
            thickness=15,     # Thinner, more elegant boxes
            line=dict(color="gray", width=0.5),
            label=all_nodes,
            color="royalblue" # Uniform node color for academic look
        ),
        link=dict(
            source=sankey_df['source'].map(node_map),
            target=sankey_df['target'].map(node_map),
            value=sankey_df['value'],
            color="rgba(173, 216, 230, 0.4)" # Light blue transparent links
        )
    )])

    fig.update_layout(
        title_text=f"Mapping the Scientific Landscape: {' > '.join(categories).
        font_size=14,
        height=700,
        margin=dict(l=50, r=50, t=80, b=50)
    )
    fig.show()

# Run it with a smaller n_items for a cleaner look
# plot_clean_sankey(df_recent, categories=['country', 'topic', 'author'], n_ite
```

General Results

Production Over Time

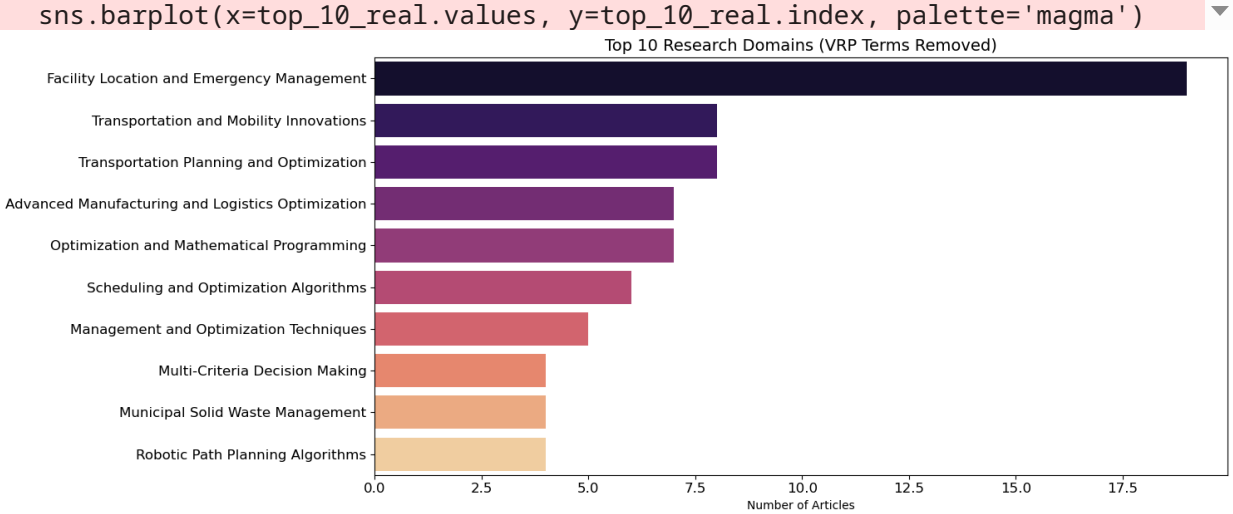
In [43]:
```
plot_production_trends(df_clean)
```



Topic Analysis

In [44]:
```
plot_real_trends(df_clean)
```

```
/tmp/ipykernel_1792766/2377348178.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_10_real.values, y=top_10_real.index, palette='magma')
```



Top Cited Articles

In [45]:
```
show_top_cited(df_clean)
```

Out[45]:

	display_name	primary_author	publication_year	cited_by_count
101	Heuristic Methods for Location-Allocation Prob...	Leon N. Cooper	1964	532
26	Location-routing problem in multimodal transpo...	Saeed Fazayeli	2018	153
12	An ant colony system empowered variable neighb...	Can B. Kalaycı	2016	141
17	Multi-objective optimization of a two-echelon ...	Alexandra Anderluh	2019	140
57	Two-echelon location-routing optimization with...	Yong Wang	2018	125
8	Deep reinforcement learning for the dynamic an...	Weixu Pan	2022	120
11	ADMM-based problem decomposition scheme for ve...	Yu Yao	2019	118
15	An innovative waste management system in a sma...	Navid Akbarpour	2021	114
22	The SR-GCWS hybrid algorithm for solving the c...	Ángel A. Juan	2009	102
14	Combination of geographical information system...	Christos D. Tarantilis	2003	91

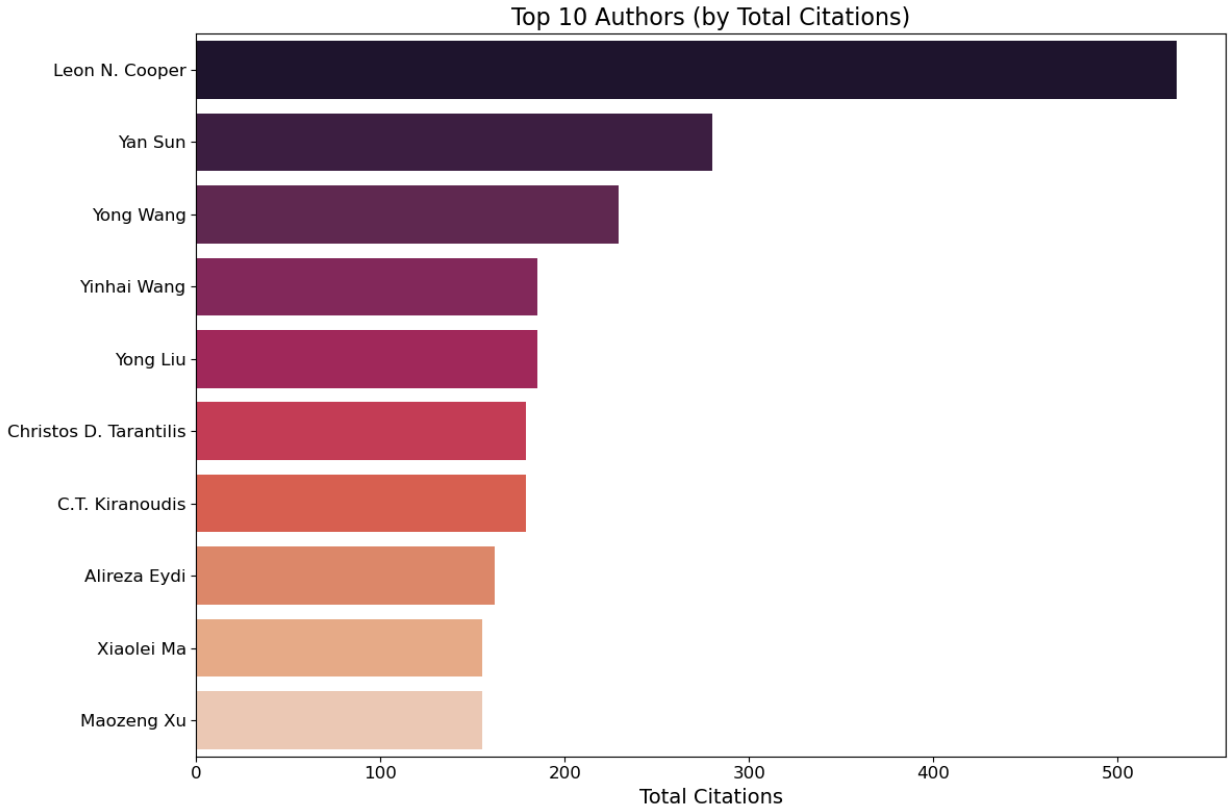
Top Authors

Per citations

In [46]:

```
plot_top_stats(df_clean,category = 'author', metric='citations')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[46]:

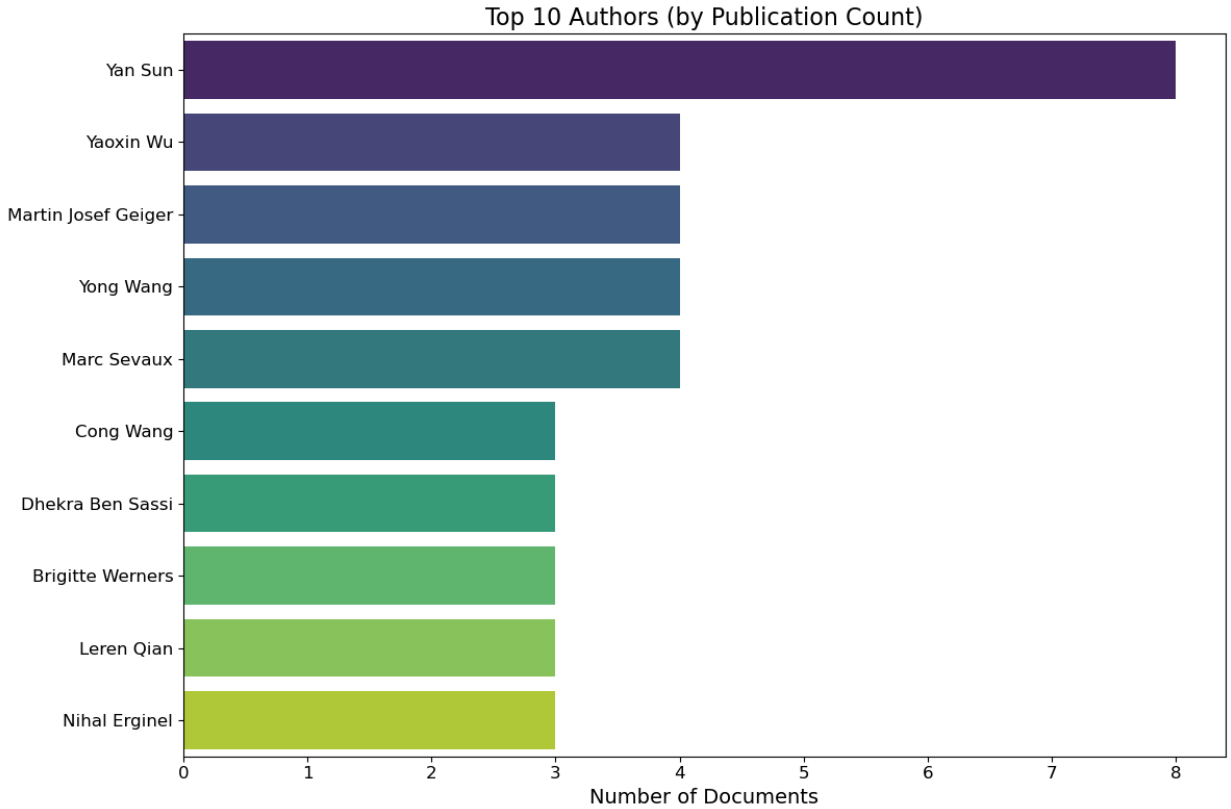
```
authorships.author.display_name
Leon N. Cooper            532
Yan Sun                   280
Yong Wang                 229
Yinhai Wang               185
Yong Liu                  185
Christos D. Tarantilis    179
C.T. Kiranoudis           179
Alireza Eydi              162
Xiaolei Ma                155
Maozeng Xu                155
Name: cited_by_count, dtype: int64
```

Per publications

In [48]:

```
plot_top_stats(df_clean,category = 'author', metric='publications')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[48]:

```
authorships.author.display_name
Yan Sun                 8
Yaoxin Wu               4
Martin Josef Geiger     4
Yong Wang               4
Marc Sevaux             4
Cong Wang               3
Dhekra Ben Sassi        3
Brigitte Werners        3
Leren Qian              3
Nihal Erginel           3
dtype: int64
```

Last 10 Years Results

In [49]:

```
df_recent = filter_by_last_n_years(df_clean, 11)

# Verify the range
print(f"Minimum year in filtered set: {df_recent['publication_year'].min()}")
```

```
Original articles: 248
Articles from 2015 to 2026: 188
Minimum year in filtered set: 2015
```

In [50]:

```
df_recent.head()
```

Out[50]:

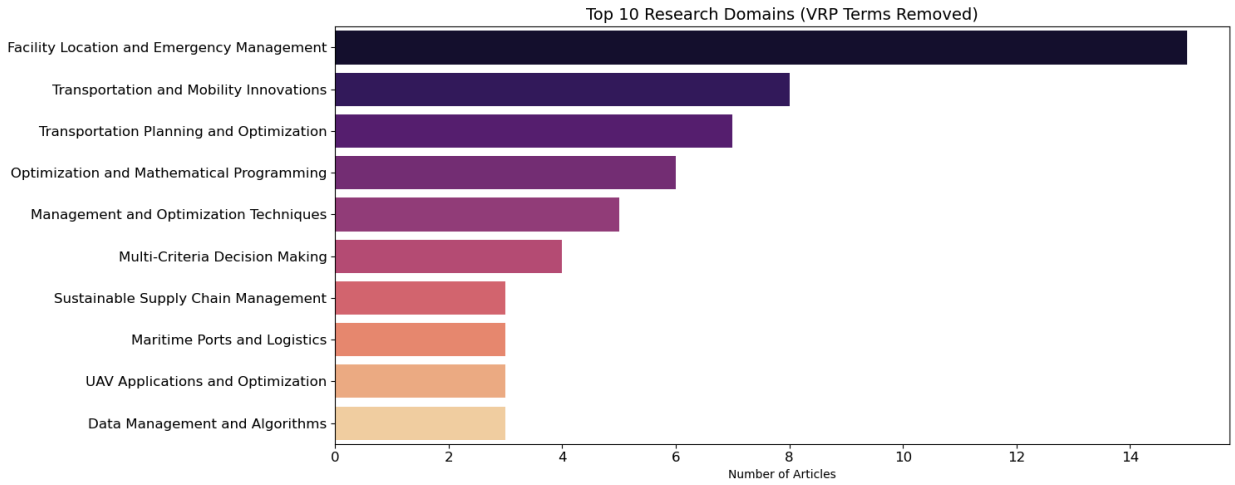
	id	abstract	authorships.author.display_name	
1	https://openalex.org/W4211132698	NaN	Merve Cengiz Toklu	https://op
2	https://openalex.org/W2999440819	NaN	Sadia Samar Ali|Turan Paksoy|Belkz Torul|Rajbi...	https://openalex.org/A507
4	https://openalex.org/W854438218	NaN	Fatemeh Torfi|Reza Zanjirani Farahani|Iraj Mah...	https://openalex.org/A500
6	https://openalex.org/W3186940976	NaN	Fatma Demircan Keskin	https://op
8	https://openalex.org/W4224138275	NaN	Weixu Pan|Shi Qiang Liu	https://openalex.org/A502

5 rows × 30 columns

Topic Analysis

In [51]:
```
plot_real_trends(df_recent)
```

```
/tmp/ipykernel_1792766/2377348178.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_10_real.values, y=top_10_real.index, palette='magma')
```



Top Cited Articles

In [52]:
```
show_top_cited(df_recent)
```

Out[52]:

	display_name	primary_author	publication_year	cited_by_count
26	Location-routing problem in multimodal transpo...	Saeed Fazayeli	2018	153
12	An ant colony system empowered variable neighb...	Can B. Kalaycı	2016	141
17	Multi-objective optimization of a two-echelon ...	Alexandra Anderluh	2019	140
57	Two-echelon location-routing optimization with...	Yong Wang	2018	125
8	Deep reinforcement learning for the dynamic an...	Weixu Pan	2022	120
11	ADMM-based problem decomposition scheme for ve...	Yu Yao	2019	118
15	An innovative waste management system in a sma...	Navid Akbarpour	2021	114
18	FH-ACO Fuzzy heuristic-based ant colony optimi...	Mohammad Shokouhifar	2021	88
43	A TimeDependent Fuzzy Programming Approach for...	Yan Sun	2018	79
2	Reverse logistics optimization of an industria...	Sadia Samar Ali	2020	77

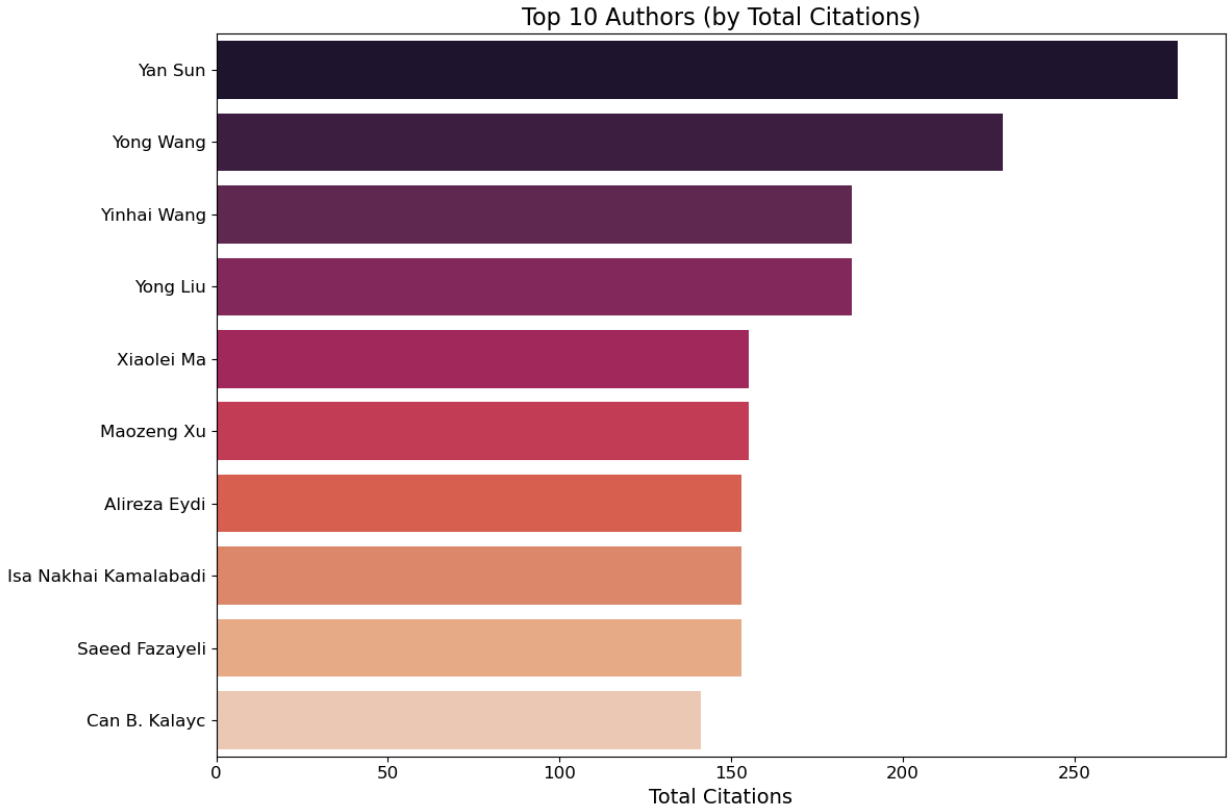
Top Authors

Per citations

In [53]:

```
plot_top_stats(df_recent,category = 'author', metric='citations')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[53]:

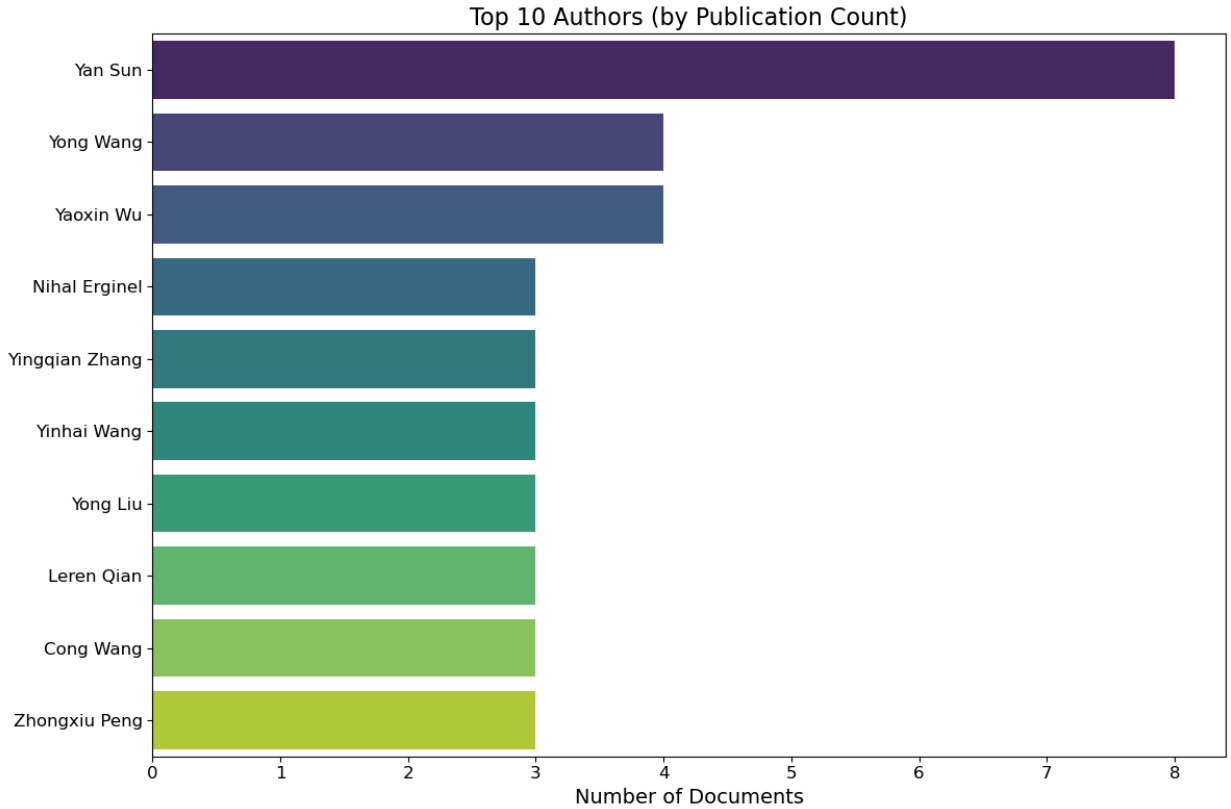
```
authorships.author.display_name
Yan Sun                    280
Yong Wang                  229
Yinhai Wang                185
Yong Liu                   185
Xiaolei Ma                 155
Maozeng Xu                 155
Alireza Eydi               153
Isa Nakhai Kamalabadi      153
Saeed Fazayeli             153
Can B. Kalayc              141
Name: cited_by_count, dtype: int64
```

Per publications

In [54]:

```
plot_top_stats(df_recent,category = 'author',metric='publications')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[54]:

```
authorships.author.display_name
Yan Sun            8
Yong Wang          4
Yaoxin Wu          4
Nihal Erginel      3
Yingqian Zhang     3
Yinhai Wang        3
Yong Liu           3
Leren Qian         3
Cong Wang          3
Zhongxiu Peng      3
dtype: int64
```

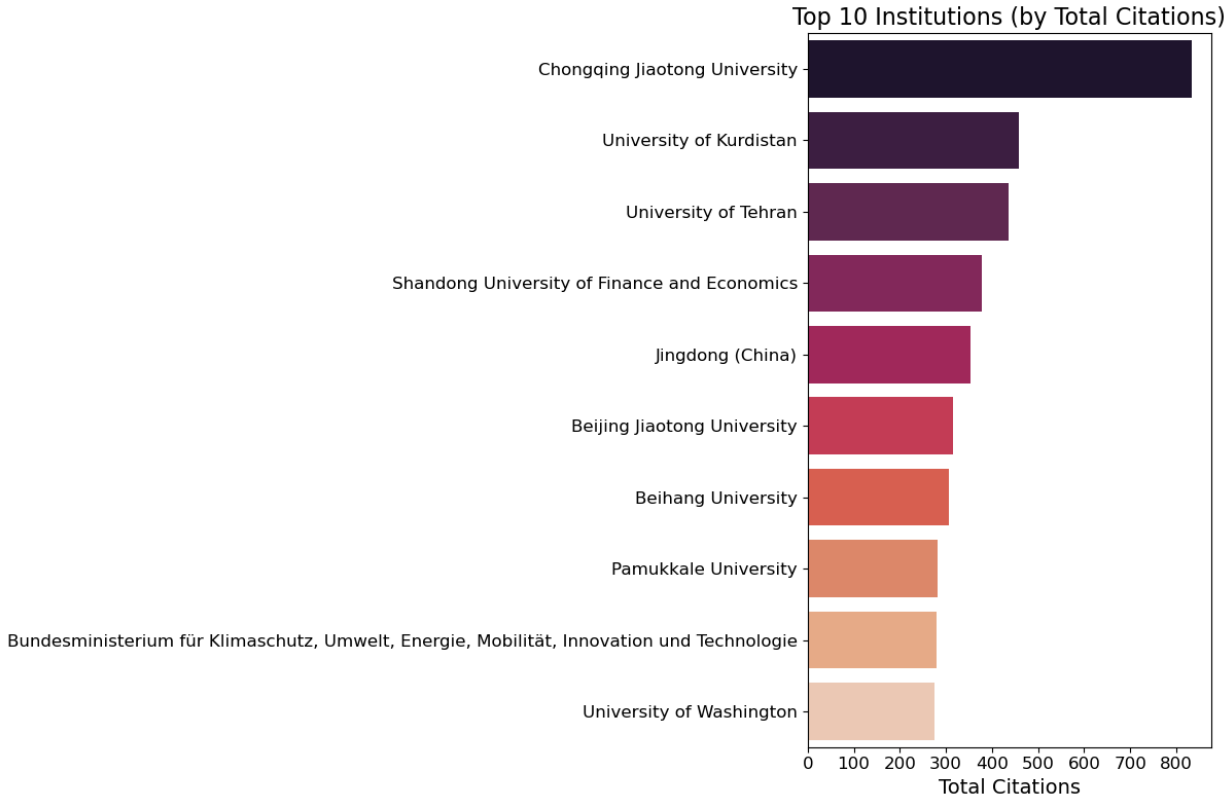
Top Institutions

Per citations

In [55]:

```
plot_top_stats(df_recent,category = 'institution', metric='citations')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[55]:

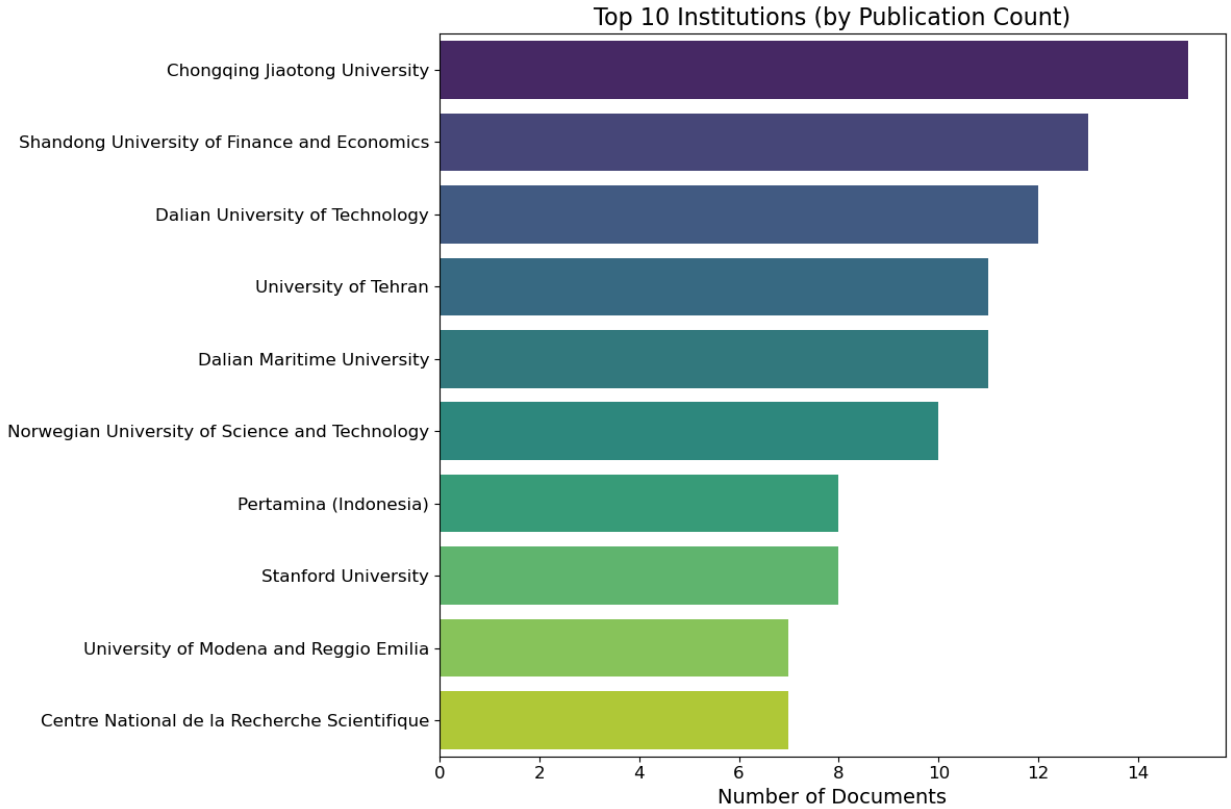
```
authorships.institutions.display_name
Chongqing Jiaotong University
834
University of Kurdistan
459
University of Tehran
435
Shandong University of Finance and Economics
379
Jingdong (China)
354
Beijing Jiaotong University
315
Beihang University
307
Pamukkale University
282
Bundesministerium für Klimaschutz, Umwelt, Energie, Mobilität, Innovation und
Technologie    280
University of Washington
275
Name: cited_by_count, dtype: int64
```

Per Publications

In [56]:

```
plot_top_stats(df_recent,category = 'institution', metric='publications')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[56]:

```
authorships.institutions.display_name
Chongqing Jiaotong University                     15
Shandong University of Finance and Economics      13
Dalian University of Technology                   12
University of Tehran                              11
Dalian Maritime University                        11
Norwegian University of Science and Technology    10
Pertamina (Indonesia)                              8
Stanford University                                8
University of Modena and Reggio Emilia             7
Centre National de la Recherche Scientifique       7
dtype: int64
```

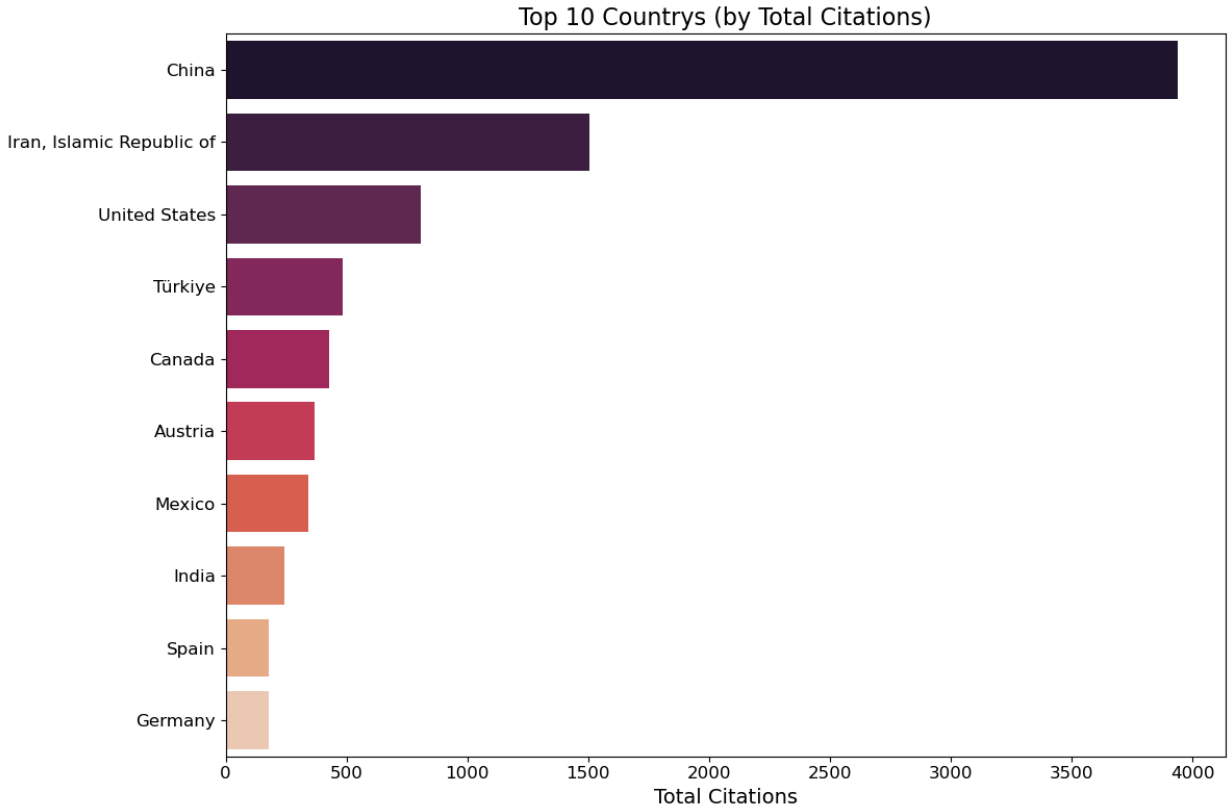
Top Countries

Per citations

In [57]:

```
plot_top_stats(df_recent,category = 'country', metric='citations')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[57]:

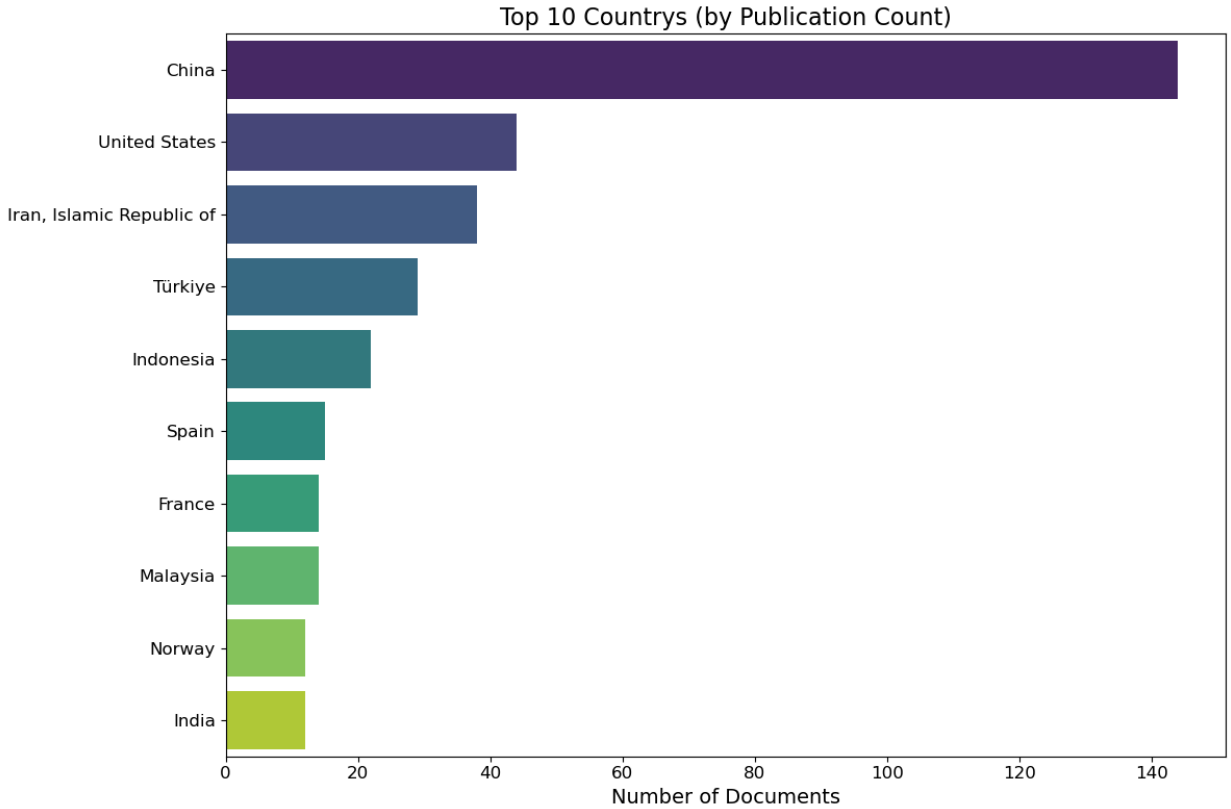
```
authorships.countries
China                        3941
Iran, Islamic Republic of    1504
United States                 809
Türkiye                       485
Canada                        430
Austria                       367
Mexico                        342
India                         244
Spain                         178
Germany                       177
Name: cited_by_count, dtype: int64
```

Per publications

In [58]:

```
plot_top_stats(df_recent,category = 'country', metric='publications')
```

```
/tmp/ipykernel_1792766/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[58]:

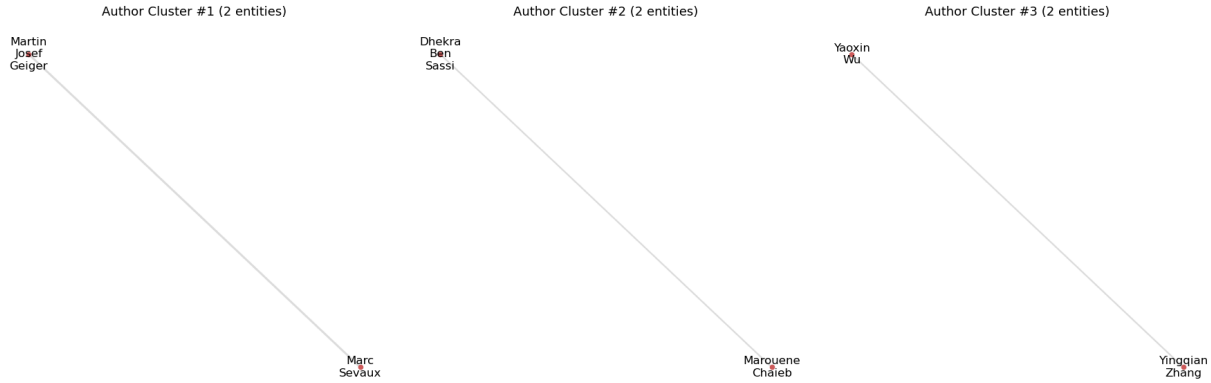
```
authorships.countries
China                        144
United States                 44
Iran, Islamic Republic of     38
Türkiye                       29
Indonesia                     22
Spain                         15
France                        14
Malaysia                      14
Norway                        12
India                         12
dtype: int64
```

Colaborations

In [63]:

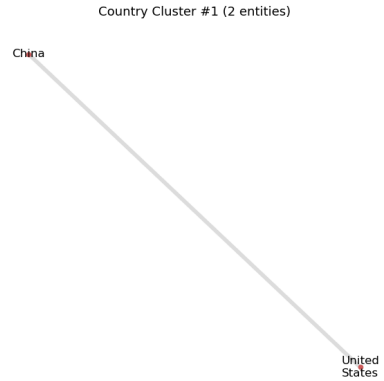
```
analyze_category_clusters(df_clean, category='author', min_shared=3, grid_rows=
```

```
Showing top 3 author clusters.
```



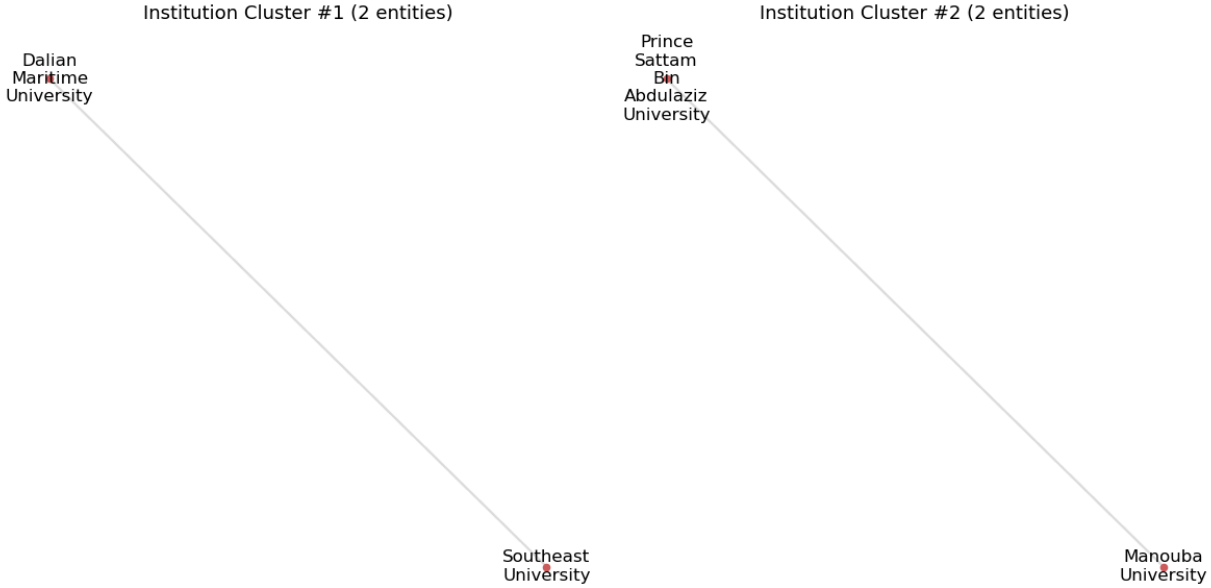
In [64]:
```
# --- Example Usages ---
analyze_category_clusters(df_clean, category='country', min_shared=5, grid_rows
```

```
Showing top 1 country clusters.
```



In [65]:
```
analyze_category_clusters(df_clean, category='institution', min_shared=3, grid_
```

```
Showing top 2 institution clusters.
```



In [ ]: